

Statistical Inference - Estimation

Statistical Inference:

Statistical inference is the branch of statistics which concerned with using probability concept to deal with certainty in decision making. The process of drawing inference about a population on the basis of information contained in a sample taken from the population is called *Statistical Inference*. Statistical inference has two parts-

- Estimation of Parameter
- Testing of Hypothesis

Estimation:

Estimation is a procedure by which we obtain an estimate of the true but unknown of a population parameter by using the sample observations $x_1, x_2, \dots, x_n$.

Example: We can estimate the mean and variance of a population by computing the mean and the variance of a sample drawn from the population.

Estimator:

Any statistic (known function of observable random variables that is itself a random variable) whose values are used to estimate $\tau(\theta)$, where $\tau(\cdot)$ is some function of parameter θ , is defined to be an estimator of $\tau(\theta)$.

Estimate:

An estimate is a numerical value of the unknown parameter obtained by applying a rule or a formula.

Example: Let $x_1, x_2, \dots, x_n$ be a random sample of size n from a population with mean μ , then $\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$ is an estimator of a mean μ and the expected value of $\bar{x}_n$ is an estimate of μ .

Show That All Estimators are Statistic but All Statistics are not Estimator.

We know that estimator by always a function of sample values by which we can estimate the parameter. Since the function of sample values are called statistic. So we can say that all estimators are statistic. But $\frac{x_i^2}{n}$ is a function of sample. So it is a statistic, we can not estimate any parameter. In this sense we can say that all estimators are statistic but all statistics are not estimator.

Distinguish Between Estimate and Estimator.

The main distinction between estimate and estimator are given below-

Estimate	Estimator
The numerical value of the estimator in a particular sample is called an estimate of the parameter.	Estimator is a statistic which is a function of sample observations and is used to estimate the value of a population parameter.
Estimate is depends on estimator.	Estimator is depends on samples.
The word estimate stands for a value of that function.	The word estimator stands for the function.
$\bar{x}_n$ is an estimate of μ .	Estimator is a variable such that $\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$ is an estimator of μ .

Properties of Estimators:

An estimator is said to be good estimator or best estimator if it has the following four properties-

1. Unbiasedness
2. Consistency
3. Efficiency
4. Sufficiency

1. Unbiasedness:

An unbiased estimator is a sample statistic for which the expected value is equal to the parameter being estimated.

An estimator t_n of the parameter θ is called unbiased for θ if their exist the relation

$$E(t_n) = \theta \quad ; \quad n=1, 2, \dots$$

Example: If $X \sim b(n, \theta)$ then $\frac{X}{n}$ the observed proportion of success is an unbiased estimator of the parameter

$$\begin{aligned} E(X) &= n\theta \\ \therefore E\left[\frac{X}{n}\right] &= \frac{1}{n}E(X) = \frac{1}{n} \cdot n\theta = \theta \end{aligned}$$

So, $\frac{X}{n}$ is an unbiased estimator of θ .

2. Consistency:

Any statistic T_n that converges in probability to a parameter θ is called a consistent estimator of that parameter θ .

$$T_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty$$

Example: Let $X_1, X_2, \dots, X_n$ be a random sample form a population with finite mean $E(X_i) = \mu < \infty$, then by Khinchine's weak law of large numbers, we have,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} E(X_i) = \mu \quad \text{as } n \rightarrow \infty$$

Hence sample mean $\bar{X}_n$ is always a consistent estimator of the population mean μ .

3. Efficiency:

A criterion which is based on the variances of the sampling distribution of estimators is usually known as efficiency.

If there are two consistent estimators T_1 and T_2 of a certain parameter θ , and if we have $\text{var}(T_1) < \text{var}(T_2)$ for all n then T_1 is more efficient than T_2 for all sample sizes.

4. Sufficiency:

An estimator is said to be sufficient for a parameter, if it contains all the information in the sample regarding the parameter. More precisely, if $T = t(x_1, x_2, \dots, x_n)$ is an estimator of a parameter θ , based on a sample $x_1, x_2, \dots, x_n$ of size n from the population with p.d.f $f(x; \theta)$ such that the conditional distribution of $x_1, x_2, \dots, x_n$ given T is independent of θ , then T is sufficient estimator for θ .

Example: Let $X_1, X_2, \dots, X_n$ be a random sample from a Bernoulli population with parameter p , then for $T = \sum X_i \sim b(n, p)$ the conditional distribution of $(x_1, x_2, \dots, x_n)$ given T is

$$\frac{f(x_1; p)f(x_2; p) \dots f(x_n; p)}{P(T = k)} = \frac{p^k q^{n-k}}{\binom{n}{k} p^k q^{n-k}} = \frac{(n-k)! k!}{n!}$$

which is independent of p . So T is the sufficient estimator for p .

Parameter Space:

The set of all admissible values of the parameters is called the parameter space or parametric space which is denoted by Ω .

Problem:

Let $x_1, x_2, \dots, x_n$ be a random sample with $E(x_i) = \mu$ and $\text{var}(x_i) = \sigma^2$ then show that $S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is a

biased estimator for σ^2 and $\hat{S}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimator for σ^2 .

Solution:

We know that

$$\begin{aligned} \sum_{i=1}^n (x_i - \bar{x})^2 &= \sum_{i=1}^n x_i^2 - n\bar{x}^2 \\ \Rightarrow E \left[\sum_{i=1}^n (x_i - \bar{x})^2 \right] &= E \left[\sum_{i=1}^n x_i^2 \right] - nE(\bar{x}^2) = \sum_{i=1}^n E(x_i^2) - nE(\bar{x}^2) \end{aligned}$$

Again,

$$\begin{aligned}\text{var}(x) &= E(x^2) - [E(x)]^2 \\ \Rightarrow E(x^2) &= \text{var}(x) + [E(x)]^2 = \sigma^2 + \mu\end{aligned}$$

$$\begin{aligned}\text{And, } \text{var}(\bar{x}) &= E(\bar{x}^2) - [E(\bar{x})]^2 \\ \Rightarrow E(\bar{x}^2) &= \text{var}(\bar{x}) + [E(\bar{x})]^2 = \frac{\sigma^2}{n} + \mu\end{aligned}$$

$$\begin{aligned}\text{Therefore, } E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right] &= \sum_{i=1}^n E(x_i^2) - nE(\bar{x}^2) = \sum_{i=1}^n (\sigma^2 + \mu) - n\left(\frac{\sigma^2}{n} + \mu\right) \\ &= n\sigma^2 + n\mu - \sigma^2 - n\mu \\ &= \sigma^2(n-1)\end{aligned}$$

$$\begin{aligned}\text{Now, } S^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \\ \Rightarrow E[S^2] &= E\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\right] = \frac{1}{n} E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right] = \frac{n-1}{n} \sigma^2\end{aligned}$$

That is why S^2 is a biased estimator of σ^2 .

Again,

$$\begin{aligned}\hat{S}^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \\ \Rightarrow E[\hat{S}^2] &= E\left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2\right] = \frac{1}{n-1} E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right] \\ &= \frac{1}{(n-1)} \cdot (n-1)\sigma^2 = \sigma^2\end{aligned}$$

That is why $\hat{S}^2$ is an unbiased estimator of σ^2 .

Problem:

Show that in random sampling from a normal population the sample mean is a consistent estimator for the population mean.

Solution:

Let $x_1, x_2, \dots, x_n$ be the random sample drawn from the normal population with mean μ and variance σ^2 .

Then the sample mean $\bar{x}$ is also normally distributed with mean μ and variance $\frac{\sigma^2}{n}$. Thus,

$$\begin{aligned}\lim_{n \rightarrow \infty} P\{|\bar{x} - \mu| > \varepsilon\} &= \lim_{n \rightarrow \infty} P\{|z| > \varepsilon\} \\ &= 1 - \lim_{n \rightarrow \infty} P\{|z| \leq \varepsilon\} \\ &= 1 - \lim_{n \rightarrow \infty} \int_{-\varepsilon}^{\varepsilon} \frac{\sqrt{n}}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{nz^2}{2\sigma^2}} dz \quad \left| \quad \text{let } t = \frac{z\sqrt{n}}{\sigma} \Rightarrow dz = \frac{\sigma}{\sqrt{n}} dt \right. \\ &= 1 - \lim_{n \rightarrow \infty} \int_{-\frac{\varepsilon\sqrt{n}}{\sigma}}^{\frac{\varepsilon\sqrt{n}}{\sigma}} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2}} dt \\ &= 1 - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2}} dt = 1 - 1 = 0\end{aligned}$$

Hence the sample mean $\bar{x}$ is a consistent estimator of the population mean μ .

Problem: If T_n is an unbiased estimator of θ , variance of T_n tends to zero as n tends to infinity, then show that T_n is consistent.

Solution:

We have to prove that T_n is a consistent estimator of θ

$$T_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty$$

$$P\{|T_n - \theta| < \varepsilon\} = 1 - \eta$$

Applying Chebyshev's inequality of the statistic T_n we get,

$$P\{|T_n - \theta_n| \leq \delta\} = 1 - \frac{\text{var}(T_n)}{\delta^2} \quad \dots \quad \dots \quad \dots \quad (1)$$

We have

$$|T_n - \theta| = |T_n - E(T_n) + E(T_n) - \theta|$$

$$\leq |T_n - E_\theta(T_n)| + |E_\theta(T_n) - \theta| \quad \dots \quad \dots \quad \dots \quad (2)$$

Now,

$$|T_n - E_\theta(T_n)| \leq \delta$$

$$\Rightarrow |T_n - \theta| \leq \delta + |E_\theta(T_n) - \theta| \quad \dots \quad \dots \quad \dots \quad (3)$$

$$\therefore P[|T_n - \theta| \leq \delta + |E_\theta(T_n) - \theta|] \geq P[|T_n - E_\theta(T_n)| \leq \delta]$$

$$\geq 1 - \frac{\text{var}(T_n)}{\delta^2} \quad \dots \quad \dots \quad \dots \quad (4)$$

Since $\theta_n \rightarrow \theta$ and $\text{var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$, there exists an n_0 such that for all $n > n_0$

$$|E_\theta(T_n) - \theta| \leq \delta_1 \quad \dots \quad \dots \quad \dots \quad (5)$$

$$\frac{\text{var}(T_n)}{\delta^2} \leq \eta \quad \dots \quad \dots \quad \dots \quad (6)$$

substituting from (5) and (6) in (4), we get,

$$P[|T_n - \theta| \leq \delta + \delta_1] \geq 1 - \eta$$

$$\Rightarrow P[|T_n - \theta| \leq \varepsilon] \geq 1 - \eta \quad \text{where } \varepsilon = \delta + \delta_1 > 0$$

$$\Rightarrow T_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty$$

So, T_n is a consistent estimator of θ .

Prove that $E(\bar{x}) = \mu$

Let the population size N and the sample size is n . Also let, $\bar{x}$ is the sample mean and μ is the population. We

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\therefore E(\bar{x}) = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{1}{n} \sum_{i=1}^n E(x_i)$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\sum_{j=1}^N x_j p(x_j) \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\sum_{j=1}^N x_j \cdot \frac{1}{N} \right] = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{N} \sum_{j=1}^N x_j \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

know that

So, sample mean $\bar{x}$ is an unbiased estimator of population mean μ .

Problem:

Let $x_1, x_2, \dots, x_n$ be a sample from Poisson distribution with parameter λ , then show that $\frac{1}{n} \sum_{i=1}^n x_i$ is an estimate

of λ . Also $\frac{2}{n(n+1)} \sum_{i=1}^n ix_i$ is an estimate of λ . So any x_i is an estimate of λ .

Solution:

Since $x_1, x_2, \dots, x_n$ be a sample from Poisson distribution with parameter λ . So, we have, $E(x) = \lambda$

Now,

$$\begin{aligned} E(\bar{x}) &= E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{1}{n} \sum_{i=1}^n E(x_i) \\ &= \frac{1}{n} \sum_{i=1}^n \lambda = \lambda \end{aligned}$$

Again,

$$\begin{aligned} E\left[\frac{2}{n(n+1)} \sum_{i=1}^n ix_i\right] &= \frac{2}{n(n+1)} E\left[\sum_{i=1}^n ix_i\right] \\ &= \frac{2}{n(n+1)} E[1 \cdot x_1 + 2 \cdot x_2 + \dots + n \cdot x_n] \\ &= \frac{2}{n(n+1)} [1 \cdot E(x_1) + 2 \cdot E(x_2) + \dots + n \cdot E(x_n)] \\ &= \frac{2}{n(n+1)} (1 \cdot \lambda + 2 \cdot \lambda + \dots + n \cdot \lambda) \\ &= \frac{2\lambda}{n(n+1)} (1 + 2 + \dots + n) \\ &= \frac{2\lambda}{n(n+1)} \cdot \frac{n(n+1)}{2} = \lambda \end{aligned}$$

$\therefore \frac{2}{n(n+1)} \sum_{i=1}^n ix_i$ is an estimate of λ . Here we can see that any x_i is an estimate of λ .

Problem:

Let $x_1, x_2, \dots, x_n$ be a random sample from $b(1, p)$, then show that $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is an estimate of p . Also show

that some other estimates of p are $T_1 = \frac{1}{2}, T_2 = x_1$ and $T_3 = \frac{x_1 + x_2}{2}$.

Solution:

Since $x_1, x_2, \dots, x_n$ be a random from $b(1, p)$, then we have, $E(x) = p$

Now,

$$\begin{aligned}
 E(\bar{x}) &= E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{1}{n} \sum_{i=1}^n E(x_i) \\
 &= \frac{1}{n} \sum_{i=1}^n p = p
 \end{aligned}$$

Again

$$E(T_1) = E\left[\frac{1}{2}\right] = \frac{1}{2} = p$$

$\therefore T_1 = \frac{1}{2}$ is an estimate of p .

$$E(T_2) = E(x_1) = p$$

$\therefore T_2 = x_1$ is an estimate of p .

$$\begin{aligned}
 E(T_3) &= E\left[\frac{x_1 + x_2}{2}\right] = \frac{1}{2}[E(x_1) + E(x_2)] \\
 &= \frac{1}{2}[p + p] = p
 \end{aligned}$$

$\therefore T_3 = \frac{x_1 + x_2}{2}$ is an estimate of p .

Problem:

Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$, then show that the $S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is a biased estimator of σ^2 . Hence also find the amount of bias.

Solution:

We have,

$$\begin{aligned}
 S^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^n [(x_i - \mu) - (\bar{x} - \mu)]^2 \\
 &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 - \frac{2}{n} \sum_{i=1}^n (x_i - \mu)(\bar{x} - \mu) + \frac{1}{n} \sum_{i=1}^n (\bar{x} - \mu)^2 \\
 &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 - \frac{2}{n} (\bar{x} - \mu)(n\bar{x} - n\mu) + (\bar{x} - \mu)^2 \\
 &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 - 2(\bar{x} - \mu)^2 + (\bar{x} - \mu)^2 \\
 &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 - (\bar{x} - \mu)^2 \\
 \therefore E(S^2) &= \frac{1}{n} \sum_{i=1}^n E(x_i - \mu)^2 - E(\bar{x} - \mu)^2 = \frac{1}{n} \sum_{i=1}^n \text{var}(x_i) - \text{var}(\bar{x}) \\
 &= \frac{1}{n} \sum_{i=1}^n \sigma^2 - \frac{\sigma^2}{n} \\
 &= \sigma^2 - \frac{\sigma^2}{n} \\
 &= \left(\frac{n-1}{n}\right) \sigma^2
 \end{aligned}$$

$\therefore S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is the biased estimate of σ^2 .

Making Unbiased:

$$\begin{aligned}
E(S^2) &= \left(\frac{n-1}{n} \right) \sigma^2 \\
\Rightarrow E\left(\frac{nS^2}{n-1} \right) &= \sigma^2 \\
\Rightarrow E\left(\frac{n}{n-1} \cdot \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right) &= \sigma^2 \\
\Rightarrow E\left(\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right) &= \sigma^2
\end{aligned}$$

So, $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimate of σ^2 .

We know that,

Amount of bias = Estimate – Parameter

$$= \left(\frac{n-1}{n} \right) \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}$$

So, $-\frac{\sigma^2}{n}$ is the amount of the bias.

Problem:

If $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimate of σ^2 then show that $E(S) \neq \sigma$ i.e. S is not an unbiased estimate of σ , where $x_1, x_2, \dots, x_n$ are iid $N(\mu, \sigma^2)$.

Solution:

Since $x_1, x_2, \dots, x_n$ are iid $N(\mu, \sigma^2)$ then we know that $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2$

Therefore,

$$\begin{aligned}
E\left[\sqrt{\frac{(n-1)S^2}{\sigma^2}} \right] &= \int_0^\infty \sqrt{x} \cdot \frac{1}{2^{\frac{n-1}{2}} \cdot \sqrt{\frac{n-1}{2}}} x^{\frac{n-1}{2}-1} e^{-\frac{x}{2}} dx \\
&= \frac{1}{2^{\frac{n-1}{2}} \cdot \sqrt{\frac{n-1}{2}}} \int_0^\infty x^{\frac{n-1}{2}} e^{-\frac{x}{2}} dx \\
&= \frac{1}{2^{\frac{n-1}{2}} \cdot \sqrt{\frac{n-1}{2}}} \cdot 2^{\frac{n}{2}} \cdot \sqrt{\frac{n}{2}} \quad \text{since } x \sim \chi_{\frac{n}{2}}^2 \\
&= \sqrt{2} \cdot \sqrt{\frac{n}{2}} \cdot \left[\sqrt{\frac{n-1}{2}} \right]^{-1} \\
E(S) &= \left[\sqrt{\frac{2}{n-1}} \cdot \sqrt{\frac{n}{2}} \cdot \left[\sqrt{\frac{n-1}{2}} \right]^{-1} \right] \cdot \sigma
\end{aligned}$$

So, S is not an unbiased estimate of σ .

random sample:

Show that unbiased estimates may not always exist.

Proof:

Consider a random variable X with p.m.f $b(1, p)$. Suppose that we wish to estimate $\Psi(p) = p^2$. Then in order that t be unbiased for p^2 , we must have,

$$\begin{aligned} E(t) &= p \cdot t(1) + (1-p) \cdot t(0) = p^2 \quad \forall p \in [0,1] \\ \Rightarrow p^2 - p[t(1) - t(0)] - t(0) &= 0 \end{aligned}$$

which is not possible because if a convergent power series $\sum_{z=0}^{\infty} c_n z^n$; $z > 0$ vanishes for all values of z lying in a closed interval the coefficients $c_0, c_1, \dots$ must be identically zero. So, unbiased estimate may not always exist.

Problem:

If $x_1, x_2, \dots, x_n$ be drawn from the population with p.d.f

$$f(x; \theta) = \begin{cases} e^{-(x-\theta)} & ; x > \theta \\ 0 & ; \text{otherwise} \end{cases}$$

then show that $(X-1)$ is an unbiased estimate of θ .

Solution:

We have,

$$\begin{aligned} E(X) &= \int_{\theta}^{\infty} x \cdot e^{-(x-\theta)} dx = 1 + \theta \\ \therefore \Rightarrow E(X-1) &= \theta \end{aligned}$$

So, $(X-1)$ is an unbiased estimate of θ .

Problem:

If $x \sim b(n, p)$ then show that $t_n(x) = \frac{x + \frac{\sqrt{n}}{2}}{n + \sqrt{n}}$ is a biased estimate of p and also find out the amount of bias. Also

show that $t_n(x)$ is asymptotically unbiased.

Solution:

We have, $E(x) = np$

Now,

$$\begin{aligned} E[t_n(x)] &= E\left[\frac{x + \frac{\sqrt{n}}{2}}{n + \sqrt{n}}\right] = \frac{E(x) + \frac{\sqrt{n}}{2}}{n + \sqrt{n}} = \frac{np + \frac{\sqrt{n}}{2}}{n + \sqrt{n}} \neq \theta \end{aligned}$$

So, $t_n(x) = \frac{x + \frac{\sqrt{n}}{2}}{n + \sqrt{n}}$ is a biased estimate of p .

We know that,

Amount of bias = Estimate – Parameter

$$\begin{aligned} &= \frac{np + \frac{\sqrt{n}}{2}}{n + \sqrt{n}} - p \\ &= \frac{np + \frac{\sqrt{n}}{2} - np - \sqrt{n}p}{n + \sqrt{n}} = \frac{(1 - 2p)\sqrt{n}}{2(n + \sqrt{n})} \end{aligned}$$

Problem:

Show that for $p(\lambda)$, there is no exactly unbiased estimator of $\frac{1}{\lambda}$.

Solution: